

EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 6A

1. Predicting with Least Squares

A courier would like to predict the time of a future delivery from its distance. The distance d_i of each delivery is known ahead of time from the route. The time t_i is what the courier measures, by timing four deliveries:

distance d_i (km)	1	2	3	4
time t_i (min)	6	5	10	11

 $\mathbf{d} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \quad \mathbf{t} = \begin{bmatrix} 6 \\ 5 \\ 10 \\ 11 \end{bmatrix}.$

- (a) First consider the model $t_i \approx s d_i$, in which time is proportional to distance. The slope s is the only unknown parameter. Write the four equations in the form $\mathbf{Ax} \approx \mathbf{b}$, identifying $\mathbf{A}$, $\mathbf{x}$ and $\mathbf{b}$. Find the least squares solution $\hat{\mathbf{x}}$ and compute the squared error $\|\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}\|^2$. What time does this model predict for a 5 km delivery?

Answer: Each observation gives one equation $s d_i \approx t_i$. Stacking them,

$$\underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} s \end{bmatrix}}_{\mathbf{x}} \approx \underbrace{\begin{bmatrix} 6 \\ 5 \\ 10 \\ 11 \end{bmatrix}}_{\mathbf{b}},$$

so $\mathbf{A} = \mathbf{d}$ is a single column, $\mathbf{x}$ is just the slope, and $\mathbf{b} = \mathbf{t}$. The least squares solution is $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$ with

$$\mathbf{A}^\top \mathbf{A} = 1 + 4 + 9 + 16 = 30, \quad \mathbf{A}^\top \mathbf{b} = 6 + 10 + 30 + 44 = 90, \quad \hat{x} = \hat{s} = \frac{90}{30} = 3.$$

The predictions and residual are

$$\mathbf{A}\hat{\mathbf{x}} = \begin{bmatrix} 3 \\ 6 \\ 9 \\ 12 \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}} = \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \quad \|\boldsymbol{\varepsilon}\|^2 = 9 + 1 + 1 + 1 = 12.$$

For a 5 km delivery the model predicts $3 \cdot 5 = 15$ minutes.

Note that s has units of minutes per km: it is $1/v$, the reciprocal of the courier's speed v . The fit $\hat{s} = 3$ min/km corresponds to $v = 1/3$ km/min, or 20 km/h.

- (b) Now consider the model $t_i \approx s d_i + c$, which adds an intercept c . Again write the four equations in the form $\mathbf{Ax} \approx \mathbf{b}$, now with $\mathbf{x} = \begin{bmatrix} s & c \end{bmatrix}^\top$. Find the least squares solution $\hat{\mathbf{x}}$, compute the squared error $\|\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}\|^2$, and predict the time of a 5 km delivery.

Compare the squared error with part (a). Which model fits the observations better, and why? Could adding the intercept ever have made the squared error larger?

Answer: Each observation now gives $sd_i + c \approx t_i$. A column of ones adds the same intercept to every prediction:

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} s \\ c \end{bmatrix}}_{\mathbf{x}} \approx \underbrace{\begin{bmatrix} 6 \\ 5 \\ 10 \\ 11 \end{bmatrix}}_{\mathbf{b}}, \quad A^\top A = \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix}, \quad A^\top \mathbf{b} = \begin{bmatrix} 90 \\ 32 \end{bmatrix}.$$

Then

$$\hat{\mathbf{x}} = (A^\top A)^{-1} A^\top \mathbf{b} = \frac{1}{20} \begin{bmatrix} 4 & -10 \\ -10 & 30 \end{bmatrix} \begin{bmatrix} 90 \\ 32 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix},$$

so $\hat{s} = 2$ and $\hat{c} = 3$. The predictions and residual are

$$A\hat{\mathbf{x}} = \begin{bmatrix} 5 \\ 7 \\ 9 \\ 11 \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \mathbf{b} - A\hat{\mathbf{x}} = \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \quad \|\boldsymbol{\varepsilon}\|^2 = 1 + 4 + 1 + 0 = 6.$$

For a 5 km delivery the model predicts $2 \cdot 5 + 3 = 13$ minutes.

The squared error dropped from 12 to 6, so the model with an intercept fits better. It has more *degrees of freedom*: two parameters to adjust instead of one. Here the intercept lets the model account for a fixed 3 minutes that every delivery takes regardless of distance, which a line through the origin cannot represent.

Adding the intercept could not have made the error larger. Setting $c = 0$ recovers the model from part (a), so every prediction available in part (a) is still available in part (b). Minimizing over a larger set of candidates can only keep the error the same or reduce it. Extra degrees of freedom can *potentially* improve the fit; they help exactly when the best intercept is not zero.

2. Least Squares: How A Shapes the Answer

Least squares finds the prediction $\mathbf{A}\hat{\mathbf{x}}$ in $\text{Col}(\mathbf{A})$ that is closest to $\mathbf{b}$. That prediction is the **projection** of $\mathbf{b}$ onto the subspace $\text{Col}(\mathbf{A})$, and the residual $\boldsymbol{\varepsilon} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$ is perpendicular to every column of $\mathbf{A}$:

$$\mathbf{A}^\top \boldsymbol{\varepsilon} = \mathbf{0}.$$

In this problem the target stays fixed while the matrix changes:

$$\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}.$$

Very little computation is needed. Try to answer each part from the geometry before computing anything.

(a) Let

$$\mathbf{A}_a = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Describe $\text{Col}(\mathbf{A}_a)$ in words. Without using the formula $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$, find the projection $\mathbf{A}_a \hat{\mathbf{x}}$, the residual $\boldsymbol{\varepsilon} = \mathbf{b} - \mathbf{A}_a \hat{\mathbf{x}}$, the squared error $\|\boldsymbol{\varepsilon}\|^2$, and the parameters $\hat{\mathbf{x}}$. Then verify that $\mathbf{A}_a^\top \boldsymbol{\varepsilon} = \mathbf{0}$.

Answer: $\text{Col}(\mathbf{A}_a)$ is the set of all vectors whose third entry is zero: the horizontal plane in $\mathbb{R}^3$. The closest point of that plane to $\mathbf{b}$ keeps the first two entries and zeroes the third:

$$\mathbf{A}_a \hat{\mathbf{x}} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}, \quad \|\boldsymbol{\varepsilon}\|^2 = 16, \quad \hat{\mathbf{x}} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

Both columns of $\mathbf{A}_a$ have third entry zero, so each has zero inner product with $\boldsymbol{\varepsilon}$, and $\mathbf{A}_a^\top \boldsymbol{\varepsilon} = \mathbf{0}$. No choice of $\mathbf{x}$ can do anything about the third entry of $\mathbf{b}$; that part is the error.

(b) Now let

$$\mathbf{A}_b = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 0 \end{bmatrix}.$$

How does $\text{Col}(\mathbf{A}_b)$ compare with $\text{Col}(\mathbf{A}_a)$? Which of the projection $\mathbf{A}\hat{\mathbf{x}}$, the residual $\boldsymbol{\varepsilon}$, the squared error $\|\boldsymbol{\varepsilon}\|^2$ and the parameters $\hat{\mathbf{x}}$ change from part (a), and which do not? Find the new $\hat{\mathbf{x}}$.

Answer: The columns of $\mathbf{A}_b$ are two linearly independent vectors in the horizontal plane, so they span the same plane: $\text{Col}(\mathbf{A}_b) = \text{Col}(\mathbf{A}_a)$. The projection depends only on the subspace, not on which vectors we use to describe it, so the projection, $\boldsymbol{\varepsilon}$ and $\|\boldsymbol{\varepsilon}\|^2 = 16$ are exactly as in part (a).

The parameters do change, because $\hat{\mathbf{x}}$ records how to build the projection from the columns we were given. Solving $x_1 + x_2 = 1$ and $x_1 - x_2 = 3$ gives

$$\hat{\mathbf{x}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad 2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}.$$

The subspace decides the prediction and the error; the columns decide the coordinates.

(c) Now let

$$\mathbf{A}_c = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

- i. Describe $\text{Col}(\mathbf{A}_c)$ and find the projection $\mathbf{A}_c \hat{\mathbf{x}}$, the residual $\boldsymbol{\varepsilon} = \mathbf{b} - \mathbf{A}_c \hat{\mathbf{x}}$ and $\|\boldsymbol{\varepsilon}\|^2$.
- ii. Before computing, could the squared error have come out smaller than in part (a)? Why or why not?
- iii. Compute $\mathbf{A}_c^\top \mathbf{A}_c$. What goes wrong with the formula $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$? Find every $\hat{\mathbf{x}}$ for which $\mathbf{A}_c \hat{\mathbf{x}}$ equals the projection you found above.

Answer:

- i. The second column is twice the first, so the columns are linearly dependent ($\mathbf{A}_c$ is *rank deficient*) and $\text{Col}(\mathbf{A}_c)$ is only the line spanned by $\mathbf{a} = [1 \ 1 \ 0]^\top$. Projecting onto that line,

$$\mathbf{A}_c \hat{\mathbf{x}} = \frac{\mathbf{a}^\top \mathbf{b}}{\mathbf{a}^\top \mathbf{a}} \mathbf{a} = \frac{4}{2} \mathbf{a} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix}, \quad \|\boldsymbol{\varepsilon}\|^2 = 18.$$

- ii. No. The line lies inside the plane from part (a), so every prediction available to $\mathbf{A}_c$ was already available to $\mathbf{A}_d$. Minimizing over fewer candidates can never give a smaller error: $18 \geq 16$.
- iii. Computing,

$$\mathbf{A}_c^\top \mathbf{A}_c = \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix}, \quad \mathbf{A}_c^\top \mathbf{b} = \begin{bmatrix} 4 \\ 8 \end{bmatrix}.$$

The second row of $\mathbf{A}_c^\top \mathbf{A}_c$ is twice the first, so it is not invertible and the formula cannot be used. We can still ask which parameters produce the projection: $\mathbf{A}_c \hat{\mathbf{x}} = [2 \ 2 \ 0]^\top$ requires $x_1 + 2x_2 = 2$, so

$$\hat{\mathbf{x}} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad \text{for any } t \in \mathbb{R}.$$

Every one of these gives the same prediction $\mathbf{A}_c \hat{\mathbf{x}}$. The projection is still unique. What is lost is uniqueness of the parameters, because the two columns point the same way and can be traded off against each other.

(d) Finally, let

$$\mathbf{A}_d = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

What is $\text{Col}(\mathbf{A}_d)$? What are the projection $\mathbf{A}_d \hat{\mathbf{x}}$ and the squared error, and is $\hat{\mathbf{x}}$ unique? What does least squares reduce to for this matrix?

Answer: The three columns of $\mathbf{A}_d$ are linearly independent, so $\text{Col}(\mathbf{A}_d) = \mathbb{R}^3$. Then $\mathbf{b}$ is already in the column space, its projection is $\mathbf{b}$ itself, and the squared error is 0. Least squares reduces to solving $\mathbf{A}_d \mathbf{x} = \mathbf{b}$ exactly. Reusing part (b) for the first two entries,

$$\hat{\mathbf{x}} = \mathbf{A}_d^{-1} \mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix},$$

and this is unique because the columns are linearly independent.